

Junior Makers · Class 5

Rain-Sensing Automatic Window

It feels the first drop and shuts itself!

Rain sensing

Water conducts electricity

Servo hinge

Did you know?

Many modern cars switch on their wipers by themselves when rain hits the windscreen.





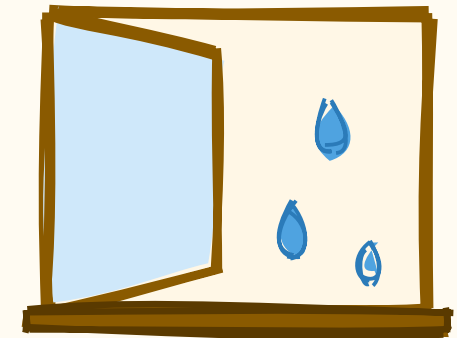
Let's think...

Monsoon rain, window open, nobody home!



Picture this

You are at school. The bedroom window at home is wide open. Suddenly, dark clouds and a heavy monsoon shower. Your books, bed and carpet are getting soaked. Who closes the window?



Phone a neighbour?

Run home?

The window itself!

Today: we build a window that feels the rain and closes itself, then warns the family.





Learning aims

Today I will learn to...



1



Explain why **water conducts electricity** and how a rain sensor uses it

2



Read a **rain sensor** and understand its numbers

3



Use a **servo** and a **hinge** to close a model window

4



Trigger an **LED** and **buzzer** alert when rain starts

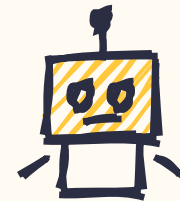
5



Choose a sensible **threshold** by testing

Keywords

conductor · rain sensor · analog ·
threshold · servo · hinge · alert



Watch and think

Video time!



Rain-Sensing Windshield Wipers

Toyota Canada · about 2 minutes · youtu.be/rscCDwS00aQ

Watch for...

- 1 Where is the rain **sensor** on the car?
- 2 What does the car do when more rain falls?
- 3 Why is this safer for the driver?



After the video: Our window will close by itself when it rains. What other machine could react to rain?



Real world

Machines that feel the rain



Car wipers

Switch on and speed up when rain hits the glass



Greenhouses

Roof vents close before plants get flooded



Smart homes

Skylights and windows shut and send a phone alert



Washing lines

Some covers roll out over clothes when it rains



In Pakistan, the monsoon (July to September) can bring **sudden heavy rain**. A window that closes in seconds could save a room.

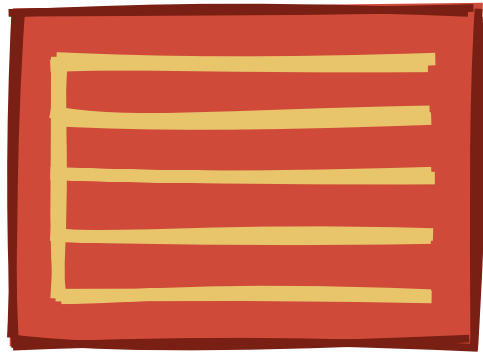
Spot it!

Ask at home: has rain ever come in through an open window?

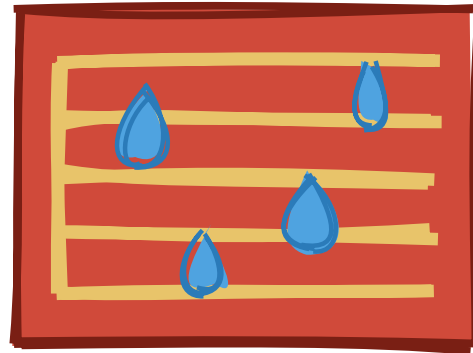


The science

Water can carry electricity

**DRY: gap between lines**

no path, almost no current

**WET: water bridges the gap**

a path! current flows

How the plate works

The plate has two sets of metal lines that **do not touch**.
A raindrop lands across two lines and makes a **bridge**.
More water = more bridges = more current.

Why rain conducts

Rainwater carries tiny bits of dissolved dust and salts. Those let electricity through. Very pure water barely conducts!

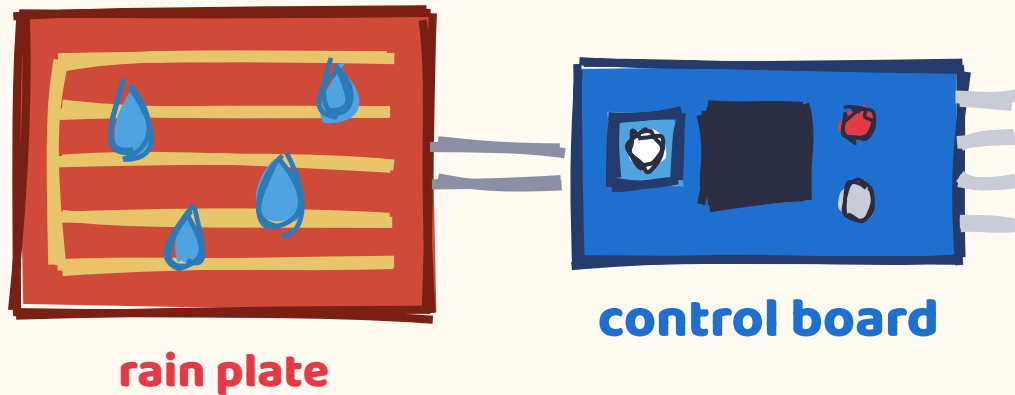


Safety: our 5V sensor is safe. Water and **mains** electricity are deadly. Never touch switches with wet hands.



Meet the sensor

The rain sensor: two parts



On the control board

Blue dial (potentiometer): sets when DO switches to "wet".

Red LED: power is on.

Green LED: lights up when DO says wet.

A small chip (LM393) compares the plate to the dial.



Pin	What it does
VCC	5V power
GND	Ground
AO	Analog out: a number 0 to 1023 (how wet)
DO	Digital out: LOW = wet, HIGH = dry

Today we use AO

A number tells us HOW wet, not just yes or no. We choose the threshold ourselves in code!



Be a scientist!

What number does rain make?

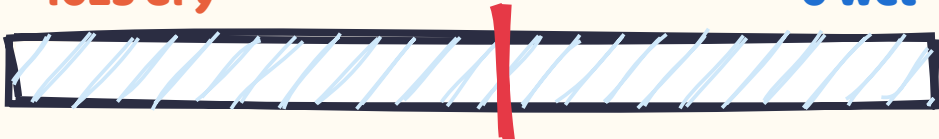


Upside-down again!

Dry plate = **big number** (near 1023).
Wet plate = **smaller number**.
Drip water on the plate with a dropper and watch the number fall.

1023 dry

0 wet



our threshold: 500?

Plate	Typical	Ours
Completely dry	about 1000	-----
2 or 3 drops	about 700	-----
Wet all over	about 400	-----
Soaked / puddle	below 300	-----

Every sensor is different
Typical numbers are only a guide. Test YOUR sensor and write YOUR numbers.





Stand up and play!

Be the rain-sensing window

How to play

1. Everyone is a window: arms **wide open**.
2. The teacher shows a card: **sun**, **drizzle** or **storm**.
3. Storm: close your arms and shout "**BEEP BEEP!**"
4. Sun: stay open. Drizzle: you decide!



Drizzle round: did everyone do the same thing? That is why the machine needs **one clear number** to decide: the **threshold**.

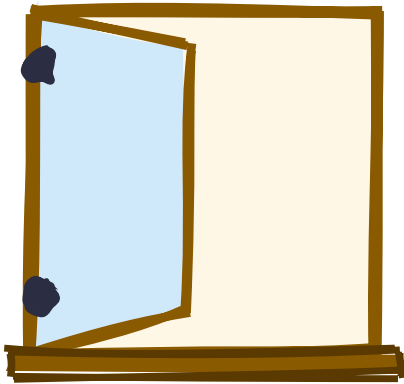
Think deeper

Should the window close for **ONE** drop? What could go wrong if it did?



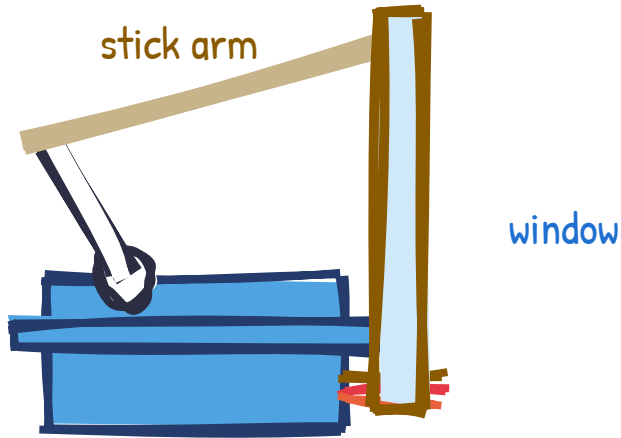
Engineering

Hinges: how the window swings



hinge on the side

A **hinge** lets a door or window turn around one edge, like your elbow.



The **servo horn** pushes a **stick arm**, and the arm pushes the window shut. This is a **linkage**.

Servo angles

0° = window open

90° = window closed

Your window may need 80° or 100°. Test it!

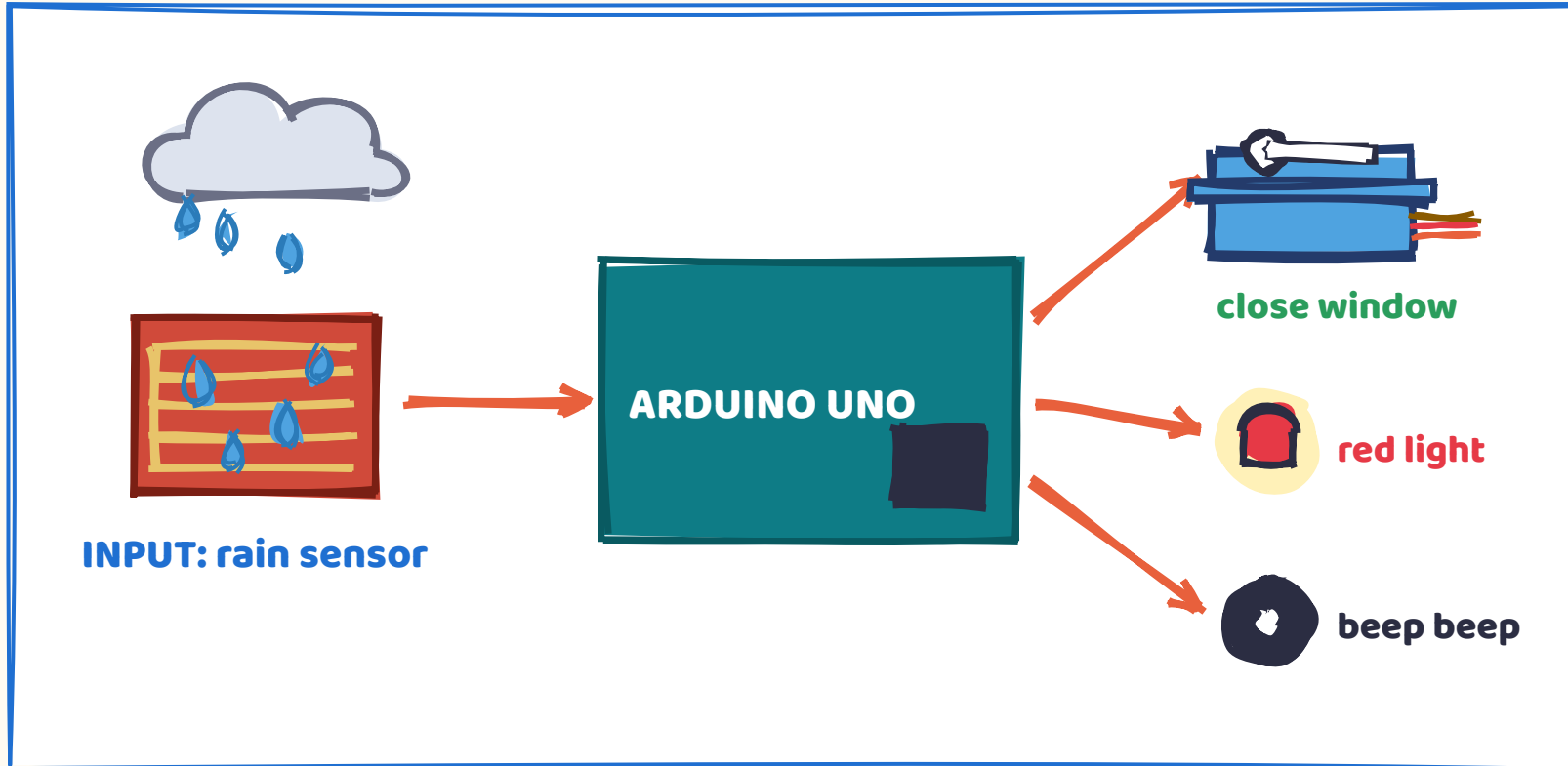
Hinge hack

Use a strip of strong tape along one edge. Bend it back and forth a few times to loosen it.



The plan

One sensor, three actions



The rule

IF rain number < 500

THEN close window, red light ON, beep.

ELSE keep window open, light OFF.

Remember?

Last class: two sensors, two jobs. Today: one sensor, three outputs!



Build it

Our window kit (per group)



Arduino Uno

The brain



Rain sensor

Plate + control board



SG90 servo

Closes the window



Red LED + 220 Ω

Rain alert light



Buzzer

Rain alert sound



Cardboard

Wall, frame, window

Also: jumper wires, breadboard, tape, ice-cream stick, **spray bottle or dropper**, tissues, a tray to catch drips.

Re-use

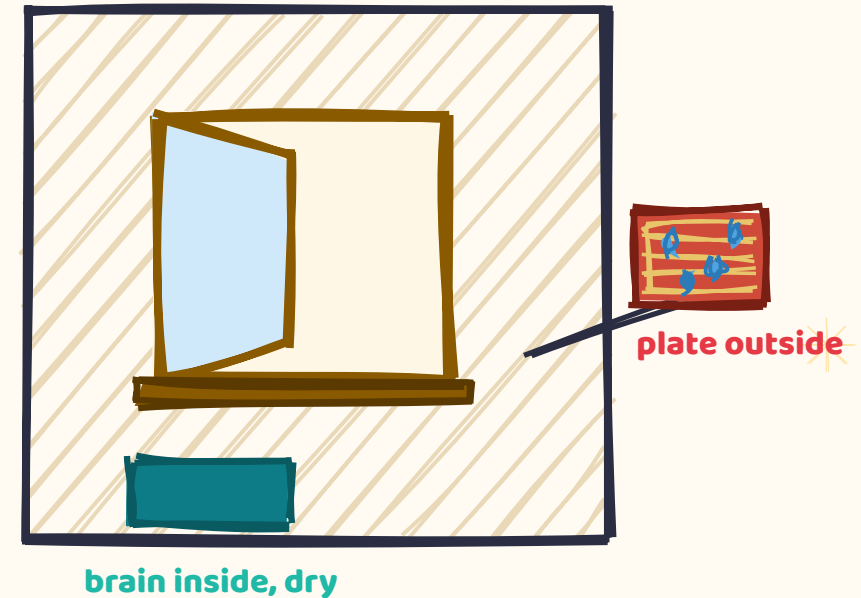
The servo and lid skills from the smart bin work here too. Engineers re-use ideas!



Build it

Make the model window

- 1 **Cut** a window hole in a cardboard wall. Keep the cut-out: that is your window.
- 2 **Hinge** the window to one side of the hole with strong tape.
- 3 **Fix** the servo inside the wall. Link the horn to the window with a stick.
- 4 **Place** the rain plate **outside**, tilted a little so water runs off.
- 5 **Keep** the Arduino and control board **inside**, far from the water.

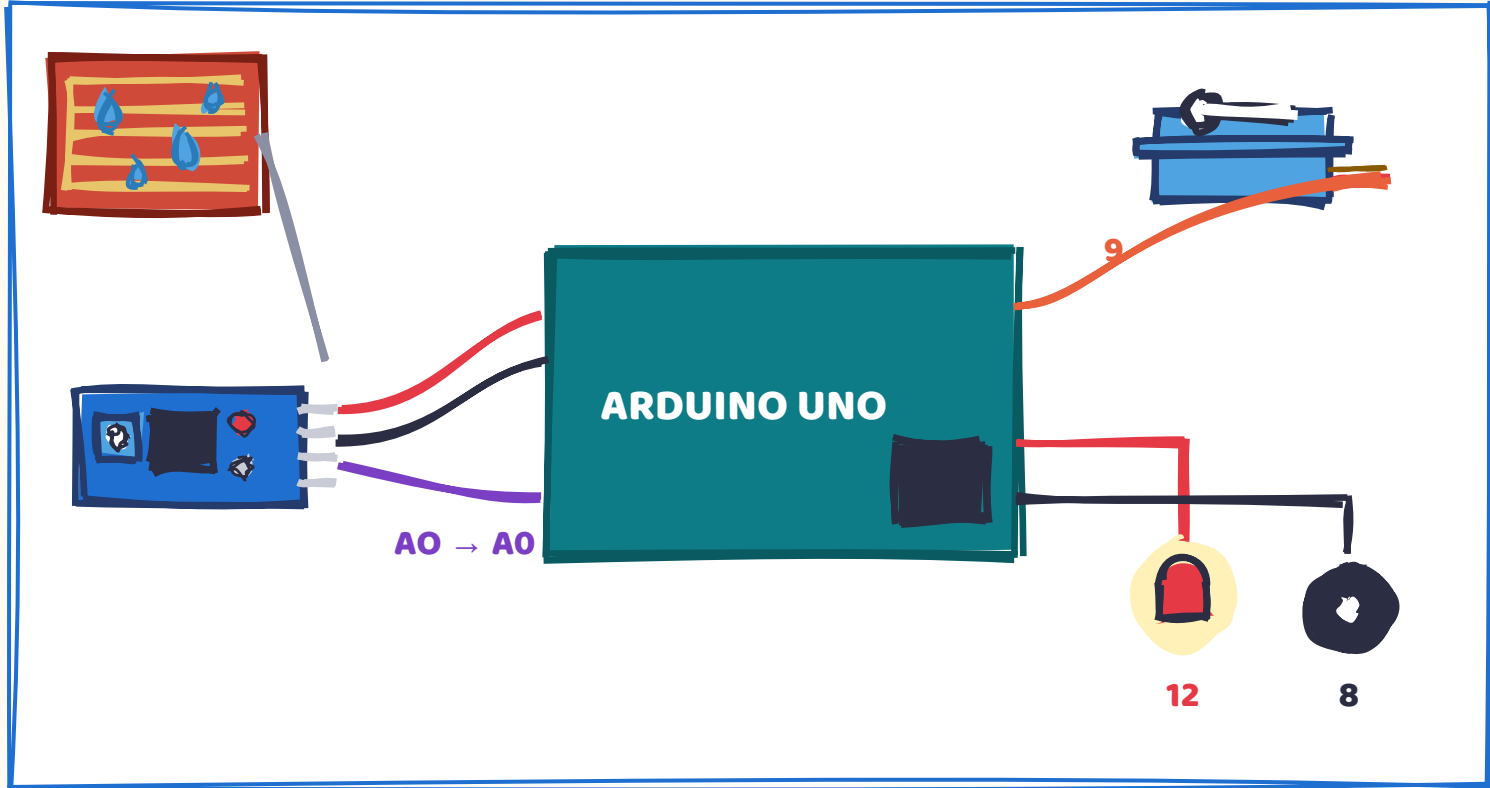


Why tilt the plate? So the water runs off and the sensor can dry, just like a real roof.



Build it

Wiring the window



Part	Arduino
Sensor VCC / GND	5V / GND
Sensor AO	A0
Servo orange	pin 9
Servo red / brown	5V / GND
Red LED (+ 220 Ω)	pin 12
Buzzer +	pin 8

Keep it dry
 Only the rain plate gets wet. The control board, breadboard and Arduino stay inside!



Code it

The rain-sensing window in ArduinoBlocks

**when Arduino starts**

servo pin 9 → angle 0 (window open)

forever

set rain to read analog A0

serial print rain

if rain < 500

servo pin 9 → angle 90 (close)

turn ON pin 12 (red light)

tone pin 8, 1000 Hz, 200 ms

else

turn OFF pin 12

wait 500 ms

Read it like a story

Start with the window **open**.

Read the rain number every half second.

IF it is wet: close, light up, beep.

ELSE: light off. The window **stays shut** until we add the challenge!



Change the 500

Use the numbers you wrote in your table. Pick a number between dry and a few drops.



Test like an engineer

Make it rain!



Test	Expect	Number	Result
Dry plate	Window open, quiet	_____	pass / fail
1 drop	?? (your threshold decides)	_____	pass / fail
Spray 3 times	Close, red light, beep	_____	pass / fail
Wipe dry with tissue	Light off. Window?	_____	pass / fail



Window does not close? Open the **Serial monitor**. Is the number really going below your threshold?



Spray rules

Spray **only** the plate, from above, like real rain.

Never spray the Arduino, breadboard or laptop.

Keep tissues and a tray ready.



Did it beep forever?

That is correct for now! The plate is still wet. Fixing it is today's challenge.





Level up!

Window challenges



Choose your challenge



Sunshine again: reopen the window when the plate is dry (number above 900).



Beep only once: beep 3 times when rain starts, then stay quiet.



Rain-o-meter: light green, yellow or red LEDs for drizzle, rain and storm.



Gentle close: close the window slowly, a few degrees at a time.

Why two numbers?

Close below **500**, open above **900**. The gap stops the window flapping open and shut while the plate is half dry.



Boss level

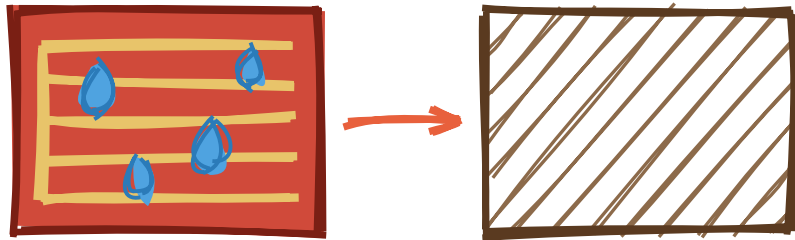
Only reopen after the plate has been dry for 10 seconds. Real rain stops and starts!





Engineer's secret

Why rain sensors wear out



new

after weeks: rusty

When electricity flows through water on metal, the metal slowly **corrodes** (rusts away).

How engineers fix it

Switch the sensor **ON** only when reading, then **OFF** again.
Wipe and dry the plate after class.
Real products use special coatings or light beams instead.

Teacher tip

Power the sensor VCC from a digital pin: turn it on, wait 10 ms, read, turn it off.



End-of-class routine: unplug USB, wipe the plate with a tissue, let it dry flat, then pack it away. Our sensors will last the whole year!





Show what you know

What did we learn?

- ☒ Rainwater conducts electricity and bridges the plate's lines
- ☒ Wetter plate = smaller number on A0
- ☒ A threshold is the number where the machine changes its mind
- ☒ A servo and a hinge make a linkage that closes the window
- ☒ One sensor can trigger many outputs: motor, light and sound



Exit ticket: the sensor reads 320. Is the window open or closed? Why?

At home

Next time it rains, look for anything that reacts to the rain by itself.



Next class: a Fire & Gas Safety Alarm Panel!

